

PROTISTS**Activity 28.1 How has endosymbiosis contributed to the diversity of organisms on Earth today?**

1. The protists have been called a “catch-all group.” What does this mean? Explain.

2. There are more than 100,000 recognized species of protists; in Campbell Biology, 11th edition, they are subdivided into four supergroups and many clades. Even though the protists are extremely diverse, we can recognize some common themes in their evolution, including the evolution of complex cell structure, novel genetic recombination strategies, and complex life cycles. In addition, this group as a whole has great ecological importance. Provide at least two examples of protists from Chapter 28 in Campbell Biology, 11th edition, that illustrate each theme. Below are a few examples that can be used to answer this question, but this is not an exhaustive list and you should come up with additional examples on your own.

Theme	Examples
a. complex cell structure	
b. novel genetic recombination strategies	
c. complex life cycles	
d. ecological importance	

3. In the late 1960s (and since), considerable evidence has been found for the endosymbiotic theory of the origin of various organelles in eukaryotic cells.

a. What is the endosymbiotic theory?

b. Which eukaryotic organelles were proposed to have arisen as endosymbionts?	c. What kinds of evidence were presented to support each organelle as an endosymbiont?

4. The chloroplasts of the Euglenoids (clade Euglenozoa) and the Chlorarachniophytes (in the clade Rhizaria) are both thought to have arisen as a result of secondary endosymbiosis.

a. What is secondary endosymbiosis?

b. What evidence supports the idea of secondary endosymbiosis?

5. Structurally, *Giardia lamblia* lacks complete mitochondria and was once thought to be an example of what the earliest living eukaryotes may have looked like.

a. If *Giardia* is similar in structure to the earliest living eukaryotes, what does this imply about the order of evolution of the various eukaryotic organelles (that is, nucleus, cytoskeleton, mitochondria, chloroplasts)?

- b. More recently, we have discovered that the Diplomonads (e.g., *Giardia*) and the Parabasalids (the other group of Excavates) have modified mitochondria.
 - i. What types of mitochondria do they have?
 - ii. What does this evidence indicate about the evolution of the various eukaryotic organelles?